

NAME: _____ PERIOD: _____ DATE: _____

Two Surveys, Two Biases

AP Statistics · Topic 3.4 (CED 1.12) · B: 2026-10-01 · E: 2026-10-09

[STAR] TODAY'S PROBLEM

Suppose a town studies support for a park fee.

Mail survey: A neutral question is mailed to 500 randomly chosen households. Only 120 return it; 78% of respondents oppose the fee.

Phone survey: The town calls 300 random phone numbers. Of 240 people who answer, 62% oppose after hearing, *Do you support a NEW FEE that would tax park users?*

One council member says 62% must be closer to the truth because more people answered. Another says 78% must be closer because the households were randomly chosen.

Park-fee surveys

	n	Reply	Opp.	Flaw
Mail	500	120	78%	Nonresp.
Phone	300	240	62%	Wording

FIRST TAKE — before the video (5 min)

Which estimate, if either, would you report to the town? Name one detail behind your choice.

[BOOK] BIAS FAVORS SOME RESPONSES (keep on the page)

Bias systematically favors some responses over others. **Nonresponse bias** is possible when selected people who do not respond differ from those who do; the response rate is responses/selected. **Response bias** can come from confusing or leading wording. A larger sample or response rate does not repair a different design flaw. Match the fix to the source of bias.

Finish after the video:

DOK 1 (a) Compute the response rate for each survey. Name the main source of bias in each.

DOK 2 (b) Explain how each bias could change the reported percent opposing the fee. State a likely direction when the information supports one.

DOK 3 (c) Decide whether either council member has established which estimate is closer to the town's true opinion. Cite the 24% response rate and the loaded wording in your reasoning, explain why neither estimate is trustworthy as is, and propose one design-specific fix for each survey.

Frame: Neither estimate is proven closer because _____.

Frame: For the mail survey, _____; for the phone survey, _____.

EXIT — turn this in

One thing the video changed about my first take: _____